

Trend Analysis of FDA Warning Letters Issued to Medical Products About Violations to Current Good Manufacturing Practices (CGMP) Between 2007 and 2014

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Therapeutic Innovation
& Regulatory Science
2016, Vol. 50(3) 312-318
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DOI: 10.1177/2168479015619201
tirs.sagepub.com

Abstract

Background: The US Food and Drug Administration (FDA) is responsible for protecting public health. Issuance of a warning letter is the FDA's principal means of achieving prompt voluntary compliance with regulations. The objective of this study is to analyze the trends of warning letters issued to medical products about current good manufacturing practices (CGMP) violations during 2007-2014, with a secondary purpose of providing suggestions to manufacturers on how to avoid warning letters. **Methods:** Warning letters were collected from the FDA website, and information about the date of issuance, company, country, and type of medical product was collected from the letters related to medical products about CGMP violations. Descriptive statistics were performed for the analysis. **Results:** A total of 997 warning letters were issued to medical products about CGMP violations during 2007-2014. It showed an upward trend in the number of warning letters from 2009 to 2011. In 2011, it reached a maximum and the FDA released 159 letters. Overall, 685 (68.7%) letters were issued to medical devices, 223 (22.4%) to finished pharmaceuticals, followed by active pharmaceutical ingredients (5.7%), biological products (including blood products) (1.8%), and pharmacy compounding (1.4%). Despite a sharp reduction in 2013, the annual number of warning letters released to medical devices was larger than others; conversely, changes in other products remained relatively stable. **Conclusions:** The issuance number of warning letters increased during 2009-2011. Medical device manufacturers received the largest numbers of letters. The issuance number and type of warning letters were greatly influenced by changes in FDA's internal enforcement procedures, drug policies and regulations, as well as other departments' regulations. Manufacturers should comply with regulations voluntarily and respond promptly to policy changes.

Keywords

Food and Drug Administration, regulatory focus, enforcement action, drug policy, medical devices excise tax

Background

The US Food and Drug Administration (FDA) was in charge of protecting the public health for more than 100 years,¹ and now it is a part of the Public Health Service under the US Department of Health and Human Services. The FDA carries the responsibility of overseeing the safety and efficacy of food, drugs, biological products, and medical devices as well as electronic products that emit radiation.²

In general, the FDA issues regulatory letters to firms or individuals as a communication channel to express the FDA's concerns about violations of the Federal Food, Drug, and Cosmetic Act (FDCA) and related regulations. Based on how serious the letters are, they can be classified into two types: untitled letters and warning letters. Untitled letters request (rather than require) a written response from the firm within a reasonable amount of time,³ while warning letters are issued for violations of regulatory significance, which are more serious than untitled

letters.⁴ A warning letter records the violative activities and requires a written response within 15 working days after receiving the letter.⁴ Furthermore, a warning letter documents that in case of failing to take prompt corrective actions, serious consequences might follow, such as seizure, injunction, or prosecution.⁴ As the most serious communication before FDA initiates an enforcement action, FDA classifies a warning letter as "informal enforcement actions"⁴ and they hope that as an

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Submitted 25-Jun-2015; accepted 02-Nov-2015

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administrative tool it could give individuals and firms an opportunity to take voluntary and effective actions to comply with the FDA regulations.^{4,5}

For companies, especially foreign manufacturing companies, receiving a warning letter means a great failure or even a disaster, because the FDA may withhold approval of requests for export certificates, or approval of pending new drug applications listing the company.⁴ Moreover, the loss of investments and sales also brings tremendous pressure on manufacturers.

Medical products are one of the main regulatory targets of the FDA. Because of the unique nature of medical products, there are a series of regulations that play an important role in the whole process from development to marketing.^{6,7} CGMP refers to the current good manufacturing practices enforced by FDA, which is the main regulatory standard for ensuring medical products consistently meet applicable requirements and specifications.^{8,9} For instance, 21 CFR 211 is the CGMP for finished pharmaceuticals and 21 CFR 600 focuses on biological products. Adherence to the CGMP regulations ensures the identity, strength, quality, and purity of the medical product. CGMP violations mean noncompliance with the related regulations and incorrect operations during the manufacturing process.⁸

The FDA conducts planned inspections of pharmaceutical manufacturing companies that provide products in the US market.¹⁰ The goals of these inspections are to determine whether the inspected firms are operating in compliance with CGMP requirements and to monitor the quality of medical products. Warning letters regarding CGMP violations record the significant violations found during inspections and could provide an insight into the trends of FDA inspections and the main deficiencies of CGMP implementation.¹¹

The majority of previous studies have focused on warning letters issued to clinical investigators and institutional review boards¹²⁻¹⁶ as well as promotional claim violations.¹⁷⁻²² To our knowledge, no studies have reviewed warning letters issued for CGMP violations based on a comprehensive evaluation of the number and type of warning letters. The FDA continues to accelerate the medical products' approval processes and strengthen their enforcement actions under the federal government's supervision, especially within the tenure of Dr Margaret Hamburg. These reforms could greatly impact the number of warning letters released.

Against this backdrop, the main objective of this study is to assess trends of warning letters issued to medical products about CGMP violations during 2007 through 2014 by summarizing the number and type of letters. In addition, providing suggestions on how to avoid a warning letter for manufacturers is the secondary purpose.

Methods

Warning letters are made available to the public under the Freedom of Information Act and can be accessed from the FDA website.²³ The "Search and Export Warning Letters to Excel"

function was used to download an Excel Spreadsheet containing all the warning letters issued from January 1, 2007, to December 31, 2014. Letters were screened manually. Warning letters related to CGMP violations about medical products were screened on the basis of "subject" and the content of the letter.

Inclusion/Exclusion Criteria

The keywords used in the primary screening were as follows: *medical device, medical devices, biological product, biological products, blood, pharmacy compounding, finished pharmaceuticals, active pharmaceutical ingredient, and active pharmaceutical ingredients*. Warning letters related to foods, tobacco products, animal products, and cosmetics were excluded from the primary screening.

Secondary screening was conducted on the basis of primary screening using the following keywords: *GMP, CGMP, good manufacturing practices, current good manufacturing practices, QSR, and quality system regulation*.

In the end, according to the content, warning letters related to GCP, GLP, and promotional claim violations were subsequently excluded. Letters with the same company name, issuing date, issuing office, and content were identified as duplicate letters and were excluded.

Data Extraction

All the warning letters screened were available on the FDA website. Warning letters released to medical products about CGMP violations were downloaded and the following information was extracted from each letter and documented in an Excel file: the date of issuance, company, country, and the type of medical products. Two researchers performed the data collection and entry. Another one verified and checked the original source if there were any discrepancies.

Data Analysis

Descriptive statistics were performed in this study.

Results

From January 1, 2007, to December 31, 2014, a total of 997 warning letters issued for violations of CGMP for medical products were reviewed. The FDA released 96 letters in 2007 and 118 in 2008. There was an upward trend in the number of letters between 2009 and 2011, which was from 113 letters in 2009 to 135 in 2010 and even up to 159 in 2011. The issuance number of warning letters in 2012 was 157, whereas there was a decline of 51 and 44 in 2013 and 2014, respectively (Figure 1).

Of the total warning letters issued regarding CGMP violations, 685 (68.7%) were related to medical devices, 18 (1.8%) were about biological products (including blood products), 14 (1.4%) focused on pharmacy compounding, and finished pharmaceuticals accounted for about 22.4% representing

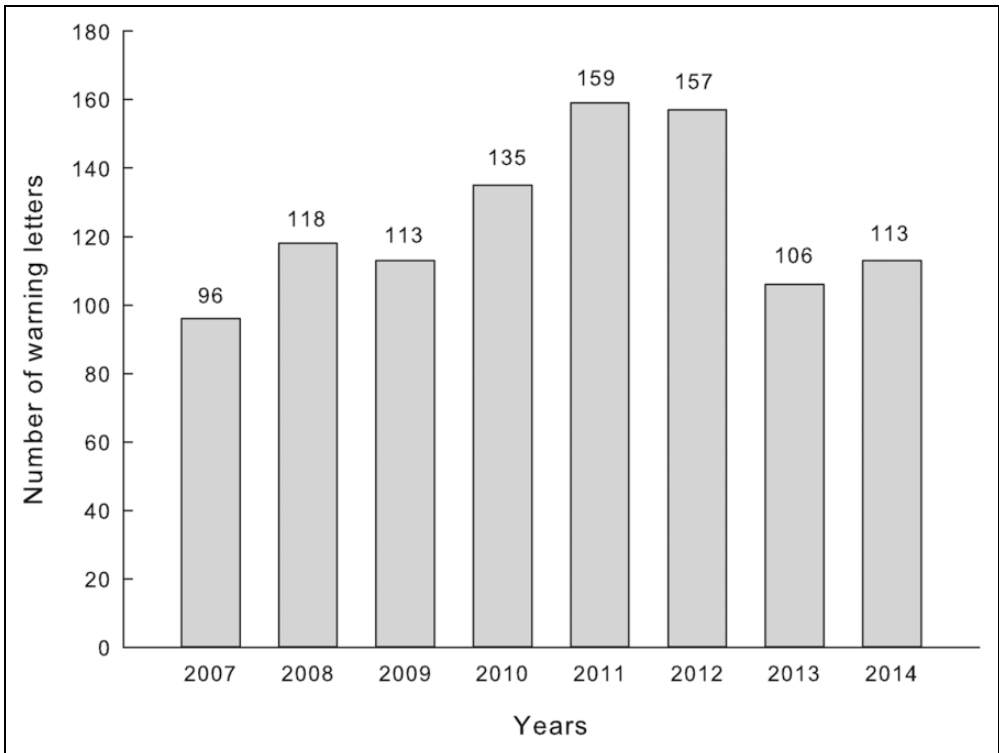


Figure 1. Warning letters related to current good manufacturing practices (CGMP) violations issued by FDA by years.

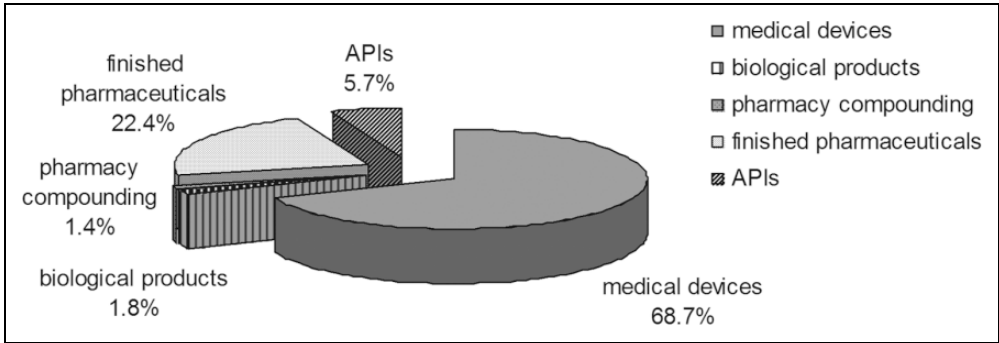


Figure 2. Percentage of warning letters about current good manufacturing practices (CGMP) violations issued to different types of medical products.

223 letters; finally, the number of letters related to active pharmaceutical ingredients (APIs) was 57 (5.7%) (Figure 2).

The analysis of the annual number of warning letters issued to different types of medical products showed differences in the issuance trends for each type of medical products. The issuance number of letters regarding medical devices was larger than others, despite there was a sharp decline from 115 in 2012 to 65 in 2013, which decreased by 43.5%. In general, the number of letters related to finished pharmaceuticals and APIs showed a small change. With respect to biological products (including blood products) and pharmacy compounding, the issuance number was small and remained relatively unchanged (Figure 3).

Discussion

This study is the first to quantitatively analyze the trends in warning letters released to medical products for CGMP violations. Medical products’ quality relates to the health and well-being of each individual. Meanwhile, this is almost always the focus of regulatory authorities and the concerns of the general public. Consumers expect that each batch of medical products they take will meet quality standards so that the products will be safe and effective. Although the impact of CGMP violations depends on the nature of those violations and the specific products involved, noncompliance with CGMP will lead to the medical products being labeled adulterated under the law.⁸ From the consumers’ point of view, adulterated

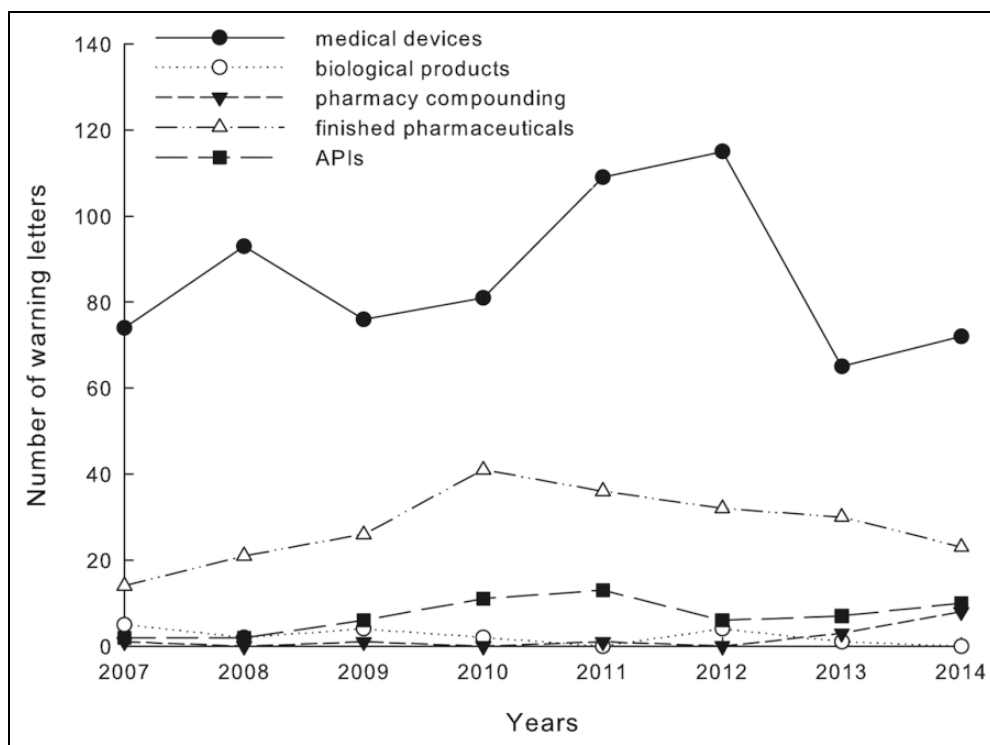


Figure 3. Annual number of warning letters about current good manufacturing practices (CGMP) violations issued to different types of medical products.

medical products are harmful to their health, and from the manufacturer's perspective, CGMP violations will cause huge losses, such as public warnings and maybe seizure of their products. Public warnings have a negative impact on corporate image, and the FDA might withhold the approval of exportation requests, or the approval of new products.⁴

A previous study conducted by Diane Nguyen et al demonstrated that compared with a substantial decline of untitled letters during 2002-2008, the number of warning letters issued by CDER and FDA district offices remained relatively unchanged, especially between 2009 and 2011, when two-thirds of the regulatory letters were warning letters.²⁴ Regulatory letters, especially warning letters, could reflect the FDA's enforcement actions and letters regarding CGMP violations could reveal policy changes in the quality management.

Results in this study showed variability in the number of warning letters regarding medical products' CGMP violations issued between 2007 and 2014. As for the total number of warning letters issued in various years, there was a spike during 2009 to 2011, which may indicate that FDA paid more attention to regulatory actions toward medical products during that period.

In 2008, FDA issued an important guidance document related to the quality of medical products for industry: Q10 Pharmaceutical Quality System. This guidance augmented CGMP by encouraging the use of science- and risk-based approaches at each lifecycle stage of medical products, which put higher requirements for industry.²⁵ The US Government

Accountability Office (GAO) highlighted the importance of FDA's supervision toward medical products in their High Risk List updated in 2009.²⁶ Also, in 2009, the FDA published a report that described four strategies to deal with the safety of medical products. These four strategies were as follows: focusing on prevention, enhancing regulatory information, improving FDA's scientific and analytic capabilities, and expanding risk-based inspection and enforcement.²⁷

Especially on August 6, 2009, FDA's former Commissioner Dr Margaret Hamburg delivered a speech which pointed out that "FDA must be vigilant, strategic, quick, and visible" and that they would implement a new enforcement strategy.²⁸ The new enforcement strategy adopted a more streamlined process to expedite the issuance of warning letters and accelerate the enforcement actions. Under the new policy, only draft letters of significant legal issues should be reviewed by the FDA's Office of Chief Counsel (OCC) rather than all the drafts of regulatory letters according to a 2002 policy.^{28,29} All of these regulatory changes might contribute to the continuous growth in the number of warning letters between 2009 and 2011.

The percentage of warning letters released to different medical products revealed FDA regulatory focus. Medical device firms received more than half of the warning letters. This situation may be attributed to several factors. On one hand, the medical device manufacturing industry has grown rapidly during the past 30 years thanks to the strong demand for medical devices in the US market.³⁰ On the other hand, with the increasingly fierce market competition, employers have tried to reduce

their labor costs by spending less time on personnel training and employing temporary workers without CGMP training.³¹ Additionally, the increasing adoption of advanced technologies has made medical devices more complex.³² The GAO repeatedly addressed the shortcoming of FDA inspections of medical devices manufacturers.³³⁻³⁶ All of these may explain why the quality management of medical devices has become the FDA's regulatory focus during the study period.

With regard to the significant reduction occurring in 2013, it may be explained by a federal excise tax act taking effect after December 31, 2012. The federal tax system imposed a 2.3% excise tax on the sale of medical devices to alleviate the increasingly serious domestic financial crisis.^{37,38} However, a status report in January 2015 from Advanced Medical Technology Association (AdvMed) showed a significant and negative impact of the medical devices excise tax based on an electronic survey of its members. The 55 responding companies accounted for 49% of total US revenues of the medical device and diagnostics industry. It said that the tax resulted in employment reductions of approximately 18,500 industry workers. Moreover, three-quarters of survey respondents have deferred or canceled capital investments and plans to open new facilities, as well as reduced investment in start-up companies because of the heavy taxation.³⁹

The results of this study suggested that biological products (including blood products) and pharmacy compounding manufacturers received fewer warning letters compared with others, which may reflect that they were not the CGMP regulatory focus during the study period. The variability in the number of warning letters released to finished pharmaceuticals and APIs was generally small. The results may reveal the mature and good quality management of these products, which has greatly benefited from years of continuous and effective regulation by the FDA. Another explanation is that it might be the result of focusing on the most significant violations of the regulation using scarce resources of the FDA.^{24,28}

In a word, the present study demonstrated that the issuance number of warning letters and regulatory focuses were strongly affected by policy. A further content analysis of the violations cited in these warning letters is needed. For companies and individuals, avoiding the receipt of warning letters not only minimizes the time, cost, and liability issues but also establishes a good image among the public.

From a manufacturer's perspective, a lot of actions can be taken to avoid FDA warning letters. First of all, paying attention to changes in the policies and regulations of the FDA and its enforcement procedures will help companies seize important opportunities of development. Then, good and timely communication with FDA and other regulatory authorities is the key to prevent misunderstanding and disagreement. For example, the new enforcement strategy in 2009 indicated that a clear, complete, and adequate response to Form FDA-483 plays an important role in avoiding the receipt of warning letters.²⁹ Third, a good understanding of the interpretations of FDA regulations is critical to achieving a high level of compliance.

Finally, a comprehensive plan to deal with FDA inspections needs to be developed. Additionally, warning letters provide insight into what is the regulatory focus of FDA and the most frequent violations observed in inspections. Therefore, it is quite important for companies to summarize the observations documented in warning letters and gain much better understanding about the regulations through the failures.

Limitations

The present study focused on the warning letters issued to medical products about CGMP violations, but not the untitled letters, which needs to be analyzed in a future study.

The exact number of firms or individuals inspected by FDA was not available on the FDA website, and only few responses to Form FDA-483 during the study period were found.

Further research is suggested for a comprehensive evaluation of all factors that could impact the FDA regulatory focus during a longer period, from the perspectives of both the pharmaceutical industry and additional federal and state departments.

Conclusions

The variability in the number of warning letters regarding CGMP violations revealed changes in FDA internal enforcement procedures and policies as well as other departments' regulations. The FDA strengthened the enforcement actions toward medical products' manufacturing process during the study period. The differences between issuance numbers of letters released to different medical products showed FDA regulatory focus. More than half of the warning letters were issued to medical device manufacturers, which indicated that the FDA is highly concerned about the quality of medical devices. Manufacturers should be sensitive to changes in FDA regulations, respond promptly to policy changes, as well as communicate efficiently with the FDA.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

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